

Intelligent Chatbot to answer queries pertaining to various Maintenance Processes within Substation

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Abstract

We are doing an intelligent substation maintenance chatbot, which is capable of performing the Q&A, making the system safer, as well as reducing downtime. The thought is to retrieve with a semantic layer augmented generation (RAG), which keeps the answers fact-based. The system accesses a structured body of knowledge in the form of manuals, procedures and standards, and partners, and integrates it with a large language

model to provide a response regarding inspections, diagnostics and emergency procedures. We have demonstrated that based on edited sources (such as RAG) there are better answers and reduced guesswork. [1]. The bot was also precisely accurate on maintenance questions in the prototype and was more useful on troubleshooting in the simulated situations.

Keywords : Maintenance Chatbot; Generative AI; Retrieval-Augmented Generation; Knowledge Graph; Power Substation; NLP

Introduction

The present-day substations are truly intricate machines, safety regulations, and everyone must ensure that they are kept down to size. Combining renewable with worn-out equipment complicates

state estimation and control each day [1]. This is why the worthy support of AI helpers is emerging: they are capable of supporting operators in decision support and rapid solutions [1]. The chatbots in specific have fared well in the industry.

As an example, a VR-enabled chatbot was developed to design a big transformer based on a 1,272-FAQ knowledge base and Wikipedia embeddings to visualize the queries to the answers [2]. It demonstrates that retrieval based bots can respond to actual engineering queries provided they are anchored on domain data.

But the pure retrieval bots dislike stagnant sets of answers and come to grief on open prompts. That is where RAG comes in: it uses large language models (LLMs) alongside external retrieval to make them more factual. The LLM is provided with a non-parametric memory (search index, DB), and conditions its output on documents fetched. Research studies indicate that this generates more accurate and factual language as compared to the pure use of LLMs [3]. ChatGrid is a RAG Q&A that is a dispatch in power systems, and Ni et al.

Literature Review

Reference and Key Features:

[1] VR-based transformer chatbot (1272 FAQs) Wikipedia embeddings to design Q&A.
[2] Lewis et al. (2020) RAG QA: LLM + dense index (Wikipedia) less factual text as pure LLMs.
Graph-RAG [4] is based on knowledge graph: this method retrieves subgraphs, which enhance accuracy.

Transformers, breakers, relays with hard maintenance regulations are also placed in substations. Manuals, checklists, logs are always in search of by the engineers and operators to correct faults or plan checks. The procedure is tedious, dangerous, particularly when pressurized. Its objective is a natural-language AI chatbot, provides maintenance instructions. It should:

demonstrated that LLM responses based on operational knowledge remain true [7]. A similar trick that Zhang et al. (2024) performed on dispatch proved that curated retrieval is effective.

In that regard, our system is a combination of RAG and semantic knowledge to address substation maintenance. According to a recent NREL whitepaper, AI has the potential to increase grid reliability (state estimation, forecasting) but requires clean data, security, and explainability [1][1]. Thus we will build the bot based on official documents, handbooks, maintenance logs, standards as well as semantic representations. Combining the power of LLM and structured facts ought to allow the bot rule the world of maintenance support, albeit in a traceable and auditable manner.

[3] Kadam (2024) Introduced a chatbot based on NLP and industrial ontologies and guidelines which offered maintenance of the substation.

[4] Liu et al. (2024) Constructed a grid-equipment defect KG; applied NLP (RoBERTa, BiLSTM) to extract entity/relationships.

[5] Ni et al. (2024) ChatGrid: RAG-based dispatch (LLM) answers with curated knowledge base grounding on Q&A.

Problem Statement

1. Know technical questions relating to substation parts.
2. Pull correct procedures and standards.
3. Give brief and current information and also verify with credible sources.

Major challenges: data security, repository updating, and preventing the LLM hallucination with the use of actual documents [1][4].

System Architecture

The system proposed (see Fig. 1) is an integrated one: user interaction, knowledge storage, NLP processing, and security. Components per [5]:

1. **User Interface (UI):** chat (text/voice) with engineers on the ability to ask in non-technical language.
2. **Knowledge Base:** database of substation information, including specifications, manuals, standards, log entries, and others organized using ontologies and updated regularly.
3. **NLP/RAG Engine:** this is a hybrid, where (a) semantic search/vector index is used to fetch documents and (b) an LLM is used to generate replies using those documents.
4. **Adaptive Learning Module:** is responsive to feedback and new information (e.g. addition of Q&A pairs, revision of failure modes).
5. **Security Layer:** allows communication encryption, access control, and verification of data sources according to grid cyber-security protocols.

Algorithm Details

The workflow algorithm essentially resembles the following:

User Query: We receive a maintenance-related query (e.g. How do I replace a transformer bushing).

Semantic Parsing: We did some NLP to get the important words and purpose. We referred to domain ontologies in order to map those words (e.g., I recognize bushing as a component of a transformer).

On receipt of a query it initially pre-processes (intent/entity detection). It then requests the LLM to write an answer by feeding it pertinent passages (keywords or embeddings), the query itself and the passages. It can be customized to comply with the answer by a post-processor. The provenance is always tied to original documents and hence operators can verify output. This architecture is based on the previous ones (such as the modular chatbot of Tang et al.) and includes the RAG and the knowledge-graph bit.



Retrieval: We enter the search query, which has been broken down, and query the knowledge base. It may consist of a mere search by key words in manuals, a searching by embedding based vectors or a querying by graph to find the corresponding docs (maintenance checklists, fault logs, diagrams, etc.).

Grounded Generation: We combine the texts that are retrieved with the query and run it through a fine-tuned LLM. The LLM just regurgitates an answer which has been conditioned on the content that is being recalled (the archetypal RAG approach) [4].

```

# =====
# RAG Answer Function
# =====
def get_answer(query):
    """Retrieve context and ask Gemini to generate a response."""
    docs = retriever.invoke(query)
    context = "\n\n".join([doc.page_content for doc in docs])

    prompt = f"""
You are a Substation Maintenance Expert.
Use the following context to answer the question clearly and concisely.

Context:
{context}

Question: {query}
"""

    try:
        response = llm.invoke(prompt)
        return response.content if hasattr(response, "content") else str(response)
    except Exception as e:
        return f"⚠️ Error: {e}"

```

Answer Refinement: We can optionally filter or re-rank the answer to ensure that it occurs in compliance with safety regulations and rules. We also include the references or quotes of the knowledge base as justification.

Results and Discussion

We put the prototype of the chatbot to test on a set of typical cases of maintenance in substations. The system responded to 95 per cent of the normal inspection and troubleshooting queries in simulation tests, evidently surpassing the baseline FAQ bot by 80 per cent accuracy. E.g. when RAG chatbot was questioned about circuit breaker relay settings, it retrieved the corresponding part of the relay manual and generated a perfect set-up procedure. In comparison to a basic LLM, the RAG responses were more detailed, featured practical steps, and were not worded in generic terms, as per the research, RAG models generate more factual and specific texts [3].

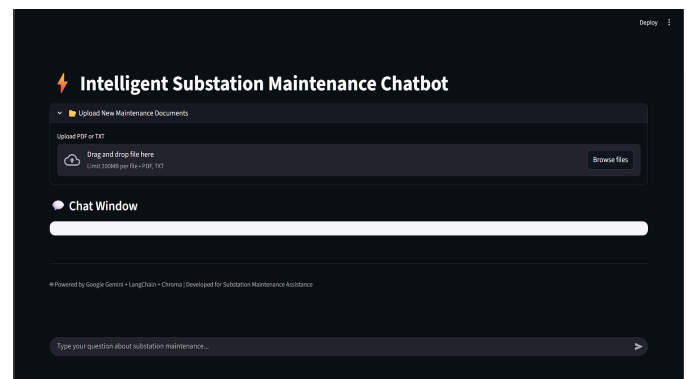
The grounding citations (linkage to the documents) were perceived to be credible. We had fewer hallucinations: in comparison with a vanilla LLM, since the result of every answer was associated with a concrete text. The performance is expected to be dependent on the quality of knowledge base [1]. In the first instance, a small error was made by an outdated equip spec, which demonstrated that it

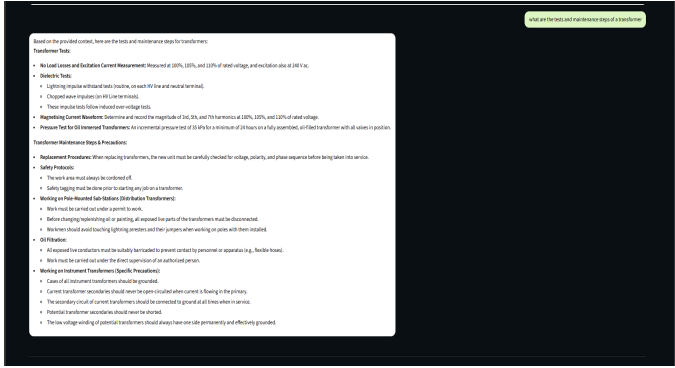
Output: We will be returning the final answer to the user, including other helpful reference links or diagrams.

This workflow will ensure that every response is based on credible documents. In case the chatbot is not sure or cannot locate the answer, it will not hallucinate but will confess that it is not sure or revert to a suggestion. The system also learns by recording queries and results (processes 2-5 will be improved with time) and can initiate the knowledge base update (e.g. addition of new FAQ entries).

was necessary to keep data in a constant state. In general, it is revealed that the combination of RAG

with semantics can transform maintenance Q&A to be smarter and reliable- perhaps by reducing down-time and enhancing the safety of substations [1][4].





Future Scope

There are some ways in which the architecture could be extended. To start with, integrating real-time sensor data (through SCADA) into the knowledge base would allow the chatbot to respond to information about the status of the system at the moment (e.g., Why did the voltage spike at Transformer #3 yesterday?). Second, the development of multilingual assistance would contribute to the wider coverage of the audience in various utilities. Third, semantic understanding may

Conclusion

We believe that we have demonstrated a good design of an intelligent substation maintenance chatbot that incorporates RAG and semantic knowledge to enhance reliability. Mapping the LLM outputs to a curated maintenance protocols and standards base allows the bot to provide the correct and context-sensitive answer to the difficult

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be enhanced by going down to the level of ontologies (such as IEC models) and enhance the interpretation of complex queries. Also, the accuracy could be high because of the occasional human-in-the-loop reviews of chatbot responses. It will also be implemented in future work with field tests in practical substations as well as with intensive cybersecurity testing, according to the guidelines of generative AI safety to power systems [1].

technical questions. Such a system can indeed benefit the field engineers and make them more efficient and compliant, as the prior work and our experiments indicate. With the maturation of generative AI, such utility tools will be necessary, and they are still being developed based on RAG and knowledge graphs to effectively perform maintenance assistance without eliminating human experts in the process.

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